

Scientific assessment and defence questions

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**Systematics studies of the continuum-coupling correlations
in near-threshold states**

Briefing for the principal opponent

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Principal assessment

- A substantial microscopic study of how the continuum reorganises nuclear resonances, decay channels and time evolution.
- The strongest result is the consistent identification of exceptional points in parameter-tuned GSM–CC Hamiltonians.
- The ^4He calculations add a valuable study of negative-parity resonances and threshold sensitivity.
- The main reservations concern quantitative validation, fitted corrections, physical interpretation of survival curves, and the definition of correlation energy.

The work supports a strong doctoral discussion; several physical claims need narrower wording or further evidence.

Source: Thesis, Introduction; Ch. 3, pp. 100–155; Conclusions, pp. 156–159.

What the thesis actually establishes

System	Main finding	Evidential status
${}^6\text{Li}, 2^+$	EP topology; finite paired strength; split peak	Strong in the tuned model
${}^7\text{Li}/{}^7\text{Be}, 5/2^-$	Mirror-channel comparison; width bifurcation	Strong diagnostics; physical relevance open
${}^8\text{Be}, 2^+$	EP and proton/ α width redistribution	Threshold and EP effects intertwined
Prepared resonances	Polynomial envelope; late-time tails	Preparation and asymptotics need checks
${}^4\text{He}$	Threshold alignment; nonrigid energy response	Clear sensitivity; interpretation open

Source: Thesis, §§3.1–3.4. “Strong” here describes support in the reported calculations, not an observed EP in a physical nucleus.

Method: a common structure and reaction framework

Gamow shell model, coupled channels

- Berggren basis: bound states, resonance poles and nonresonant continuum.
- Antisymmetrised cluster channels and outgoing boundary conditions.
- Complex poles $z = E_r - i\Gamma/2$ connect spectra and decay.

Two implementations

- Inert ^4He core for lithium and beryllium; effective interactions and channel corrections.
- No-core GSM-CC for ^4He , with HO internal states and continuum relative motion.
- Thresholds, channel truncations and calibration choices matter in both.

Major strength: structure, scattering and decay are treated in a shared many-body framework.

Source: Thesis, Ch. 2, pp. 18–69; model definitions, pp. 103, 116–117, 134, 138–140.

Exceptional points: complementary diagnostics

- Two complex eigenvalues and their eigenvectors coalesce.
- Local splitting has square-root structure:

$$z_{\pm} - z_{\text{EP}} \sim \pm \sqrt{a \delta V_p + b \delta V_n}.$$

- Phase rigidity tends to zero.
- A loop around the ${}^6\text{Li}$ EP exchanges eigenvalue sheets.

Why this is persuasive

Agreement among eigenvalue, eigenvector and topological tests is much stronger than a near-degeneracy alone. **What remains to quantify**

Numerical tolerance, contour and channel-space stability, and sensitivity to alternative acceptable interactions.

Source: Thesis, pp. 79–94; Figs. 3.3–3.5, 3.13–3.17, 3.33–3.34.

The EPs found by varying two spin–orbit strengths

System	J^π	$V_{\text{SO}}^{\ell=1}(p)$ MeV	$V_{\text{SO}}^{\ell=1}(n)$ MeV	Γ_{EP} keV
${}^6\text{Li}$	2^+	8.463	7.801	111
${}^7\text{Li}$	$5/2^-$	5.67	4.97	$\simeq 477$
${}^7\text{Be}$	$5/2^-$	4.38	4.48	477
${}^8\text{Be}$	2^+	2.2640	1.9116	$\simeq 99$

These are locations in **Hamiltonian parameter space**. They are not laboratory control settings or measured degeneracies.

The nearly equal mirror widths are a numerical result after separate retuning, rather than a protected mirror-symmetry identity.

Source: Thesis, pp. 105–106, 117–120, 136. ${}^7\text{Li}$ is also quoted as 476 keV elsewhere.

${}^6\text{Li}$: a clear microscopic EP demonstration

- The starting 2^+ poles have $(E, \Gamma) = (0.234 \text{ MeV}, 168 \text{ keV})$ and $(1.282 \text{ MeV}, 95 \text{ keV})$.
- At the EP: $E = 0.564 \text{ MeV}$ and $\Gamma = 111 \text{ keV}$; energies are relative to the ${}^4\text{He}$ core.
- Loop exchange, square-root trajectories and vanishing rigidity support the identification.
- Individual spectroscopic factors diverge with opposing contributions; their paired sum stays finite.
- Calculated elastic scattering has a split peak; intermediate-time EP survival has a polynomial prefactor.

Source: Thesis, pp. 105–114; Figs. 3.3–3.7 and 3.10.

${}^6\text{Li}$: physical calibration is the weak point

Widths at the starting Hamiltonian

	Experiment	GSM-CC
	keV	keV
2_1^+	1300	168
2_2^+	541	95

Underestimated by factors 7.7 and 5.7.

Width ratio: 2.40 versus 1.77.

Distance to the EP

The starting spin-orbit strengths are 3.492 MeV for protons and 3.936 MeV for neutrons.

The EP requires increases by factors 2.42 and 1.98.

Assessment

Robust EP evidence in the continued model; quantitatively realistic signatures for physical ${}^6\text{Li}$ are less established.

Source: Thesis, Table 3.1, p. 103; Fig. 3.2, p. 104; discussion, p. 106. Experimental values as quoted by the thesis.

${}^7\text{Li}$ and ${}^7\text{Be}$: the mirror comparison

	${}^7\text{Li}$	${}^7\text{Be}$
Doublet	$5/2_1^-, 5/2_2^-$	$5/2_1^-, 5/2_2^-$
Open channels at EP	$\alpha + t$	$\alpha + {}^3\text{He}$ and $p + {}^6\text{Li}$
Pole at EP	$E = -4.023 \text{ MeV}$, $\Gamma \simeq 477 \text{ keV}$	$E = -2.36 \text{ MeV}$, $\Gamma = 477 \text{ keV}$
Scattering response	Clear split peak in the single open channel	Dominant cluster channel retains a weaker split peak; proton response suppressed

At the ${}^7\text{Be}$ EP, the ${}^3\text{He}$ partial width is approximately ten times the proton partial width.

Open-channel composition controls how the same pole singularity appears in different reactions.

Source: Thesis, pp. 117–127; Figs. 3.13–3.17 and 3.22–3.23. Energies use the model's Hamiltonian origin.

Mirror nuclei: singular pole strengths, regular subspace

- Individual complex spectroscopic factors show singular real and imaginary parts as the two states coalesce.
- Their summed contributions remain finite, removing the singularity associated with separating the two pole states.
- Electromagnetic $B(E2)$ quantities exhibit related singular behaviour near the EP.
- This is valuable evidence of the breakdown of an isolated-state description.

A physical response must combine the coalescing subspace, interference and continuum; a divergent pole strength is not an infinite measured transition rate.

Source: Thesis, pp. 121–125; Figs. 3.18–3.21.

Width bifurcation and channel alignment

- Trajectories connecting two EPs develop a broad and a trapped resonance.
- In ${}^7\text{Li}$, the narrow width approaches zero between the EPs.
- In ${}^7\text{Be}$, the narrow state becomes proton-decaying while the broad state aligns with the ${}^3\text{He}$ channel.
- The narrow ${}^7\text{Be}$ state still has a finite proton width: it decouples from one channel, not the entire continuum.
- Total-width conservation is not exact along these scans, since the spin–orbit variation also moves energies relative to thresholds.

Source: Thesis, pp. 127–130; Figs. 3.24–3.28. The second EPs are in nonphysical parameter regions.

Mirror nuclei: validation and experimental interpretation

Width / keV	Experiment	Model
${}^7\text{Li } 5/2_1^-$	880	729
${}^7\text{Li } 5/2_2^-$	89	51
${}^7\text{Be } 5/2_1^-$	1200	691
${}^7\text{Be } 5/2_2^-$	400	179

Especially for ${}^7\text{Be}$, these are appreciable discrepancies.

- Cluster corrections differ between mirrors; no uncertainty band for EP coordinates is provided.
- A split peak or a total 2π phase evolution can also arise from ordinary interfering resonances.
- A convincing experimental claim needs pole extraction, backgrounds, channel information and uncertainties.

Source: Thesis, Figs. 3.12 and 3.15, pp. 117, 120; model corrections, pp. 117, 119; scattering discussion, pp. 125–127.

^8Be : an EP and competing decay partitions

- The well-known 2^+ doublet near 16.6 and 16.9 MeV excitation is described with α , proton, neutron and deuteron partitions.
- At $V_{\text{SO}}(p) = 2.2640$ MeV and $V_{\text{SO}}(n) = 1.9116$ MeV, the poles coalesce with $\Gamma \simeq 99$ keV.
- Phase rigidity remains close to unity over much of the scan, then collapses sharply near the EP.
- Opening $p + ^7\text{Li}$ redistributes width between proton and α decay while the total width changes little.

The new value is the microscopic multichannel account of the doublet, rather than the first general EP interpretation of it.

Source: Thesis, pp. 134–138; Figs. 3.32–3.35; earlier EP treatment acknowledged on pp. 135–136.

⁸Be: strengths and unresolved model dependence

Strengths

- Better starting width scale than in ⁶Li: 98/97 keV versus experimental 108/74 keV.
- Partial-width tracking gives direct reaction-channel interpretation.
- The sharp rigidity collapse provides a useful contrast to the mirror cases.

Weaknesses / checks

- Complex channel corrections are phenomenological:

$$c(2^+) = 1.0160 + i0.0016.$$

- Their effect on flux conservation and pole locations needs explanation.
- Threshold-driven redistribution needs a non-EP comparison.
- Channel notation and identical- α exchange symmetry deserve clarification.

Source: Thesis, p. 134; Fig. 3.32, p. 135; Figs. 3.34–3.35, p. 137.

EP time evolution: a meaningful but conditional result

For a second-order Jordan block, $H = z_{\text{EP}}I + N$ with $N^2 = 0$:

$$e^{-iHt/\hbar} = e^{-iz_{\text{EP}}t/\hbar} \left(I - \frac{iNt}{\hbar} \right).$$

For a specified preparation and projection, the pole contribution is

$$A_{\text{pole}}(t) = (a + bt)e^{-iz_{\text{EP}}t/\hbar}, \quad S_{\text{pole}}(t) = |a + bt|^2 e^{-\Gamma_{\text{EP}}t/\hbar}.$$

- The thesis finds a clear intermediate-time polynomial envelope in ${}^6\text{Li}$ and ${}^7\text{Li}$; the ${}^7\text{Be}$ effect is much weaker.
- The coefficient b depends on the prepared state and its projection onto the nilpotent part.
- The number of open channels alone does not fix the visibility of the quadratic term.

Source: Thesis, pp. 91–94, 109–115, 130–133; Figs. 3.10, 3.29–3.31. Formula includes explicit $\hbar$.

Survival calculations: preparation and positivity

What is shown

- Cutoff radii 10, 20 and 60 fm give different reconstructed spectra and late-time strengths.
- Broad short-, intermediate- and late-time regimes persist.
- Some reconstructed spectral functions become negative.

A quantitative preparation plateau and a real-energy cross-check would strengthen the time-domain conclusions.

Source: Thesis, Eqs. 2.142, 2.158–2.160; pp. 109–112, Figs. 3.8–3.9. Wang et al. (2023), [Sec. II](#).

What must be established

- For physical unitary evolution,

$$\rho_{\Psi}(E) = \sum_c |\langle E, c | \Psi \rangle|^2 \geq 0.$$

- Is the plotted function this density or a projected non-Hermitian response?
- Normalising $A(0)$ alone does not establish physical evolution for all t .

Long-time tails: which threshold sets the exponent?

For a neutral two-body threshold with $\rho(E) \propto (E - E_{\text{th}})^{L+1/2}$:

$$A(t) \sim t^{-(L+3/2)}, \quad S(t) \sim t^{-(2L+3)}.$$

- L is the **asymptotic relative angular momentum of the fragments**, not an internal valence-orbit label.
- The thesis associates t^{-5} tails with predominantly internal p waves.
- The physical ${}^6\text{Li } 2^+$ channel $\alpha + d$ has $L = 2$; ${}^7\text{Li } 5/2^-$ $\alpha + t$ has $L = 3$.
- Both channels are charged: Coulomb changes the threshold analysis, so the neutral formula cannot simply be substituted.

Ask for the computed spectral edge and a converged local exponent; a finite-time t^{-5} fit is not yet an asymptotic derivation.

Source: Thesis, pp. 72–78, 112, 130; Figs. 3.11, 3.29. Wang et al. (2023), [Eq. \(5\)](#), [Coulomb qualification](#).

^4He : the no-core extension

- All four nucleons are active, with $p + t$, $n + ^3\text{He}$ and $d + d$ reaction partitions.
- Internal fragment configurations use a harmonic-oscillator basis; relative motion uses a 90-point Berggren discretisation.
- Increasing spaces from $8\hbar\Omega$ to $16\hbar\Omega$ reduce changes: the last step gives up to 20 keV in energies and 8% in widths.
- This is evidence of convergence *within the adopted channel construction*; target and deuteron truncations remain.
- The NN interaction, regulator/transformation and treatment of three-body forces need an unambiguous specification.

Source: Thesis, pp. 138–142; Fig. 3.36. HO length $b = 1.8$ fm; deuteron CM restriction $N_{\text{CM}} \leq 2$.

^4He : separate calibration from prediction

Already calibrated	Fragment thresholds are placed at experiment: $p + t = -8.482$, $n + ^3\text{He} = -7.718$, $d + d = -4.449$ MeV.
Further fitted	J^π -dependent factors reproduce five positions: 0_2^+ , 0_1^- , 2_1^- , 2_2^- , 1_1^- .
Outside this fit	Ground-state energy, 1_2^- position, widths and branching fractions.
Large correction	In the 2^- sector, $c_d = 0.39674$ while $c_{\text{int}} = 1.06703$.

The corrected ground state is -25.415 MeV versus -28.296 MeV experimentally: 2.881 MeV underbinding.

The unfitted 1_2^- lies 534 keV below its experimental position relative to $p + t$.

Widths are unfitted observables, but their improvement partly follows from fitting decay energies.

Source: Thesis, Tables 3.3–3.5, pp. 142–144; qualification about energy-dependent widths, p. 154.

^4He : widths improve, but discrepancies remain

State	Experiment	Raw ($16\hbar\Omega$)	Corrected ($14\hbar\Omega$)
Width / MeV			
0_2^+	0.50	0.578	0.792
0_1^-	0.84	6.010	1.356
2_1^-	2.01	3.875	2.725
2_2^-	5.01	2.225	2.723
1_1^-	6.20	4.289	4.336
1_2^-	6.10	5.947	5.725

- The 0_1^- improvement is substantial; broad widths are not uniformly reproduced.
- The 2_2^- discrepancy is a factor 1.84, larger than the text's stated maximum of 1.6.

Source: Thesis, Tables 3.3 and 3.5, pp. 142, 144. Experimental comparison uses the thesis's quoted values.

Threshold alignment: useful structure diagnostics

- Shifting the thresholds strongly changes the channel content of nearby negative-parity states.
- The 1_2^- proton-channel weight reaches about 0.93 just below the neutron threshold.
- For 0_1^- , neutron weight rises from roughly 0.54 to 0.74 as its threshold is approached.
- These reorganisations accompany changes in partial widths and dispersive energy response.

Interpretation limit

Weights are real parts of Gamow c -product norms. They can be negative and are not ordinary positive probabilities or direct observables.

Source: Thesis, Eqs. 3.3–3.4, pp. 147–150; Figs. 3.37–3.38.

Threshold scans and the meaning of correlation energy

- Simultaneous threshold shifts give a nonrigid energy response.
- Reported well depths are about 1.5–2 MeV for 0_1^- , and 1–1.5 MeV for 1_1^- and 2_1^- .
- Minima cluster near 1.6–1.75 MeV relative to $p + t$, about 0.9 MeV above the neutron threshold.
- This establishes sizeable threshold sensitivity, but the scan does not supply a unique continuum-off reference.

To call a well depth a continuum correlation energy, define the varied operator, reference Hamiltonian and subtraction.

Source: Thesis, pp. 150–154; Fig. 3.39.

The missing deuteron branch is an informative failure

1_2^- decay

- Experimental $d + d$ branching: about 3%.
- Raw partial width: 0.04 keV against a total width near 5.9 MeV.
- Corrected pole is 131 keV below the $d + d$ threshold, closing that channel.
- Its deuteron weight remains negligible even in threshold scans.

Implications

- The failure is not explained solely by the corrected threshold position.
- Target excitation, cluster configurations and deuteron CM space are candidate limitations.
- Three discarded deuteron-associated states require explicit norm/CM diagnostics, since they also persist as N grows.

Source: Thesis, pp. 140–146, 150; Fig. 3.36; Table 3.6.

A definite algebraic correction: the double-pole cross section

For the ideal single-channel factor, let $\Delta = E - E_r$ and

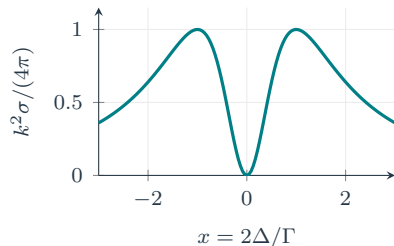
$$S_{\text{EP}} = \left(\frac{\Delta - i\Gamma/2}{\Delta + i\Gamma/2} \right)^2.$$

Direct evaluation gives

$$\sigma_{\text{el}} = \frac{\pi}{k^2} |S_{\text{EP}} - 1|^2 = \frac{4\pi}{k^2} \frac{\Gamma^2 \Delta^2}{(\Delta^2 + \Gamma^2/4)^2}.$$

Eq. 2.216 instead has a minus sign in one denominator factor. The printed expression can become negative or singular.

Source: Thesis, Eqs. 2.212–2.216, pp. 94–96; here using lower-half-plane poles. The curve is an analytic illustration, not thesis data.



Analytic illustration: constant width, one channel, no background.

Corrections and qualifications to raise proportionately

Eq. 2.210, p. 94	With residues c_0, c_1 defined in the resolvent, the pole amplitude is $(c_0 - ic_1 t/\hbar)e^{-izt/\hbar}$. A redefinition of c_1 must be explicit.
Fig. 3.37 / p. 150	The plotted $d+d$ weight is roughly $+0.05$ to -0.02 , not within 2×10^{-3} of zero.
Table 3.5 / p. 144	Upper 2^- width differs by a factor 1.84; the stated maximum is 1.6. The promised markers of fitted levels are absent.
Channel lists	${}^4\text{H} \rightarrow {}^4\text{He}$ and $\alpha {}^3G_4 \rightarrow {}^1G_4$ on p. 134; 3P_3 is impossible for $L = S = 1$ (pp. 116, 134).
Physical wording	Qualify “probability”, “unique experimental fingerprint”, “decoupled from the continuum”, and claims that general R -matrix descriptions cannot represent the physics.

Printed errors, numerical validation gaps and unsettled physical interpretations are different levels of concern.

Source: Thesis, locations above; see also pp. 7, 126, 129, 147–154, 157–158.

Overall strengths and defence priorities

Scientific strengths

- Consistent microscopic treatment of open nuclear systems.
- Several complementary EP diagnostics.
- Instructive mirror and channel comparisons.
- Explicit partial widths and model-space study.
- Reported failures reveal concrete limitations.

Priority discussion

- Candidate's individual contribution and novelty.
- Robustness under fits, truncations and numerical changes.
- Observability at the physical Hamiltonian.
- Preparation, positivity and charged-channel tails.
- Calibrated versus predicted ^4He quantities.

Connect the model diagnostics to finite, reproducible nuclear observables and decisive experiments.

Source: Synthesis of Chs. 2–4. Assessment is based on the manuscript; original code and experimental data were not reanalysed.

Sources and reading conventions

- **Primary source:** David Cardona Ochoa, *Systematics studies of the continuum-coupling correlations in near-threshold states*, Université de Caen Normandie, 11 August 2026; supplied final PDF.
- Page references use **printed thesis pages**: Arabic page n is PDF page $n + 14$. Figure and table labels refer to the thesis.
- Cardona Ochoa, Płoszajczak and Michel, *Gamow shell model description of exceptional point in ${}^7\text{Li}$* , EPJ Web Conf. **342**, 01023 (2025). [arXiv:2510.01928](#).
- Cardona Ochoa et al., *Double pole S -matrix singularity in the continuum of ${}^7\text{Be}$* , Acta Phys. Pol. B Proc. Suppl. **19**, 1-A1 (2026). [arXiv:2510.27293](#).
- Wang, Nazarewicz, Volya and Ma, *Probing the Non-exponential Decay Regime in Open Quantum Systems*, Phys. Rev. Research **5**, 023183 (2023). [Full text](#).
- Descouvemont and Baye, *The R -matrix theory*, Rep. Prog. Phys. **73**, 036301 (2010). [arXiv:1001.0678](#).

Defence questions 1–2: contribution and EP identification

1. What is your most important original scientific contribution?

Which result goes beyond previously known EP behaviour or earlier GSM–CC work, and which parts of the method and implementation are specifically yours?

2. How do you establish defectiveness, rather than a close degeneracy?

Explain how eigenvector coalescence, vanishing phase rigidity, square-root splitting and loop exchange support the EP claim. Which test is decisive numerically?

Source: Q1: Introduction, pp. 1–8; Conclusions, pp. 156–159. Q2: pp. 79–94; Figs. 3.3–3.5, 3.13–3.17.

Defence questions 3–4: robustness and physical relevance

3. What uncertainty would you attach to the EP locations?

How do the coordinates and predicted signatures change under continuum-contour refinement, channel-space enlargement and alternative fits of comparable quality?

4. What does the tuned ${}^6\text{Li}$ EP tell us about the physical nucleus?

The EP requires roughly doubling the spin–orbit strengths while the starting widths are much too small. Which conclusions survive that distance from the calibrated Hamiltonian?

Source: Q3: model definitions and EP scans, pp. 103–138. Q4: Table 3.1, p. 103; Figs. 3.2–3.4, pp. 104–106.

Defence questions 5–6: finite responses and observability

5. What physical quantity replaces a divergent pole strength?

Near the EP, individual spectroscopic factors and $B(E2)$ quantities become singular. How would you construct a finite measurable response from the coalescing states and continuum?

6. What measurement distinguishes an EP from two interfering poles?

Are a split peak and a 2π phase evolution sufficient? State the conditions behind the ideal double-pole formula and explain the role of Coulomb background and other channels.

Source: Q5: pp. 106–108, 121–125; Figs. 3.6, 3.18–3.21. Q6: pp. 94–99, 125–127; Eq. 2.216; Figs. 3.22–3.23.

Defence questions 7–8: mirror channels and decay dynamics

7. What drives the mirror-channel alignment and width bifurcation?

Separate the roles of Coulomb, thresholds and independently fitted couplings. Why can the trapped ${}^7\text{Be}$ state retain proton decay while losing its ${}^3\text{He}$ width?

8. What physical preparation is represented by your survival curves?

Explain the cutoff packet, negative reconstructed spectra and the polynomial EP envelope. What validates positivity and the evolution near a defective Hamiltonian?

Source: Q7: pp. 117–130; Figs. 3.23–3.28. Q8: Eqs. 2.142, 2.158–2.160, 2.210; Figs. 3.8–3.10, 3.29–3.31.

Defence questions 9–10: asymptotics and no-core calibration

9. Which threshold produces the reported t^{-5} tail?

The open $\alpha + d$ and $\alpha + t$ channels have relative $L = 2$ and $L = 3$ and contain Coulomb. Can you derive the tail from the calculated threshold density and demonstrate its asymptotic convergence?

10. What is predictive in the corrected ${}^4\text{He}$ calculation?

Specify the NN interaction and any three-body terms. After placing thresholds and fitting five energies, how do you assess unfitted widths and the large 2^- deuteron correction?

Source: Q9: pp. 72–78, 112, 130; Figs. 3.11, 3.29; Wang et al. (2023), Eq. (5). Q10: pp. 139–145; Tables 3.3–3.6.

Defence questions 11–12: omitted channels and correlation energy

11. How are spurious states distinguished from missing cluster physics?

What norm and centre-of-mass diagnostics justify discarding the deuteron-associated states? Could target excitations or the $N_{\text{CM}} \leq 2$ restriction explain the missing 1_2^- deuteron branch?

12. How would you define and test the continuum correlation energy?

In the threshold scan, what operator changes and what is the reference? Propose a controlled calculation that isolates the continuum contribution and an observable that could validate it.

Source: Q11: pp. 65–69, 140–146, 150; Fig. 3.36; Table 3.6. Q12: pp. 147–154; Figs. 3.37–3.41.